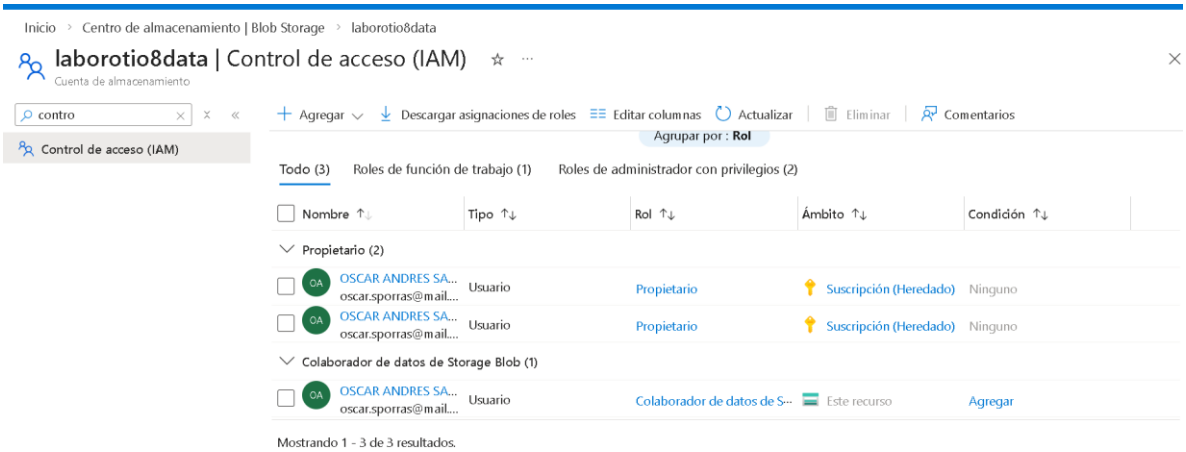
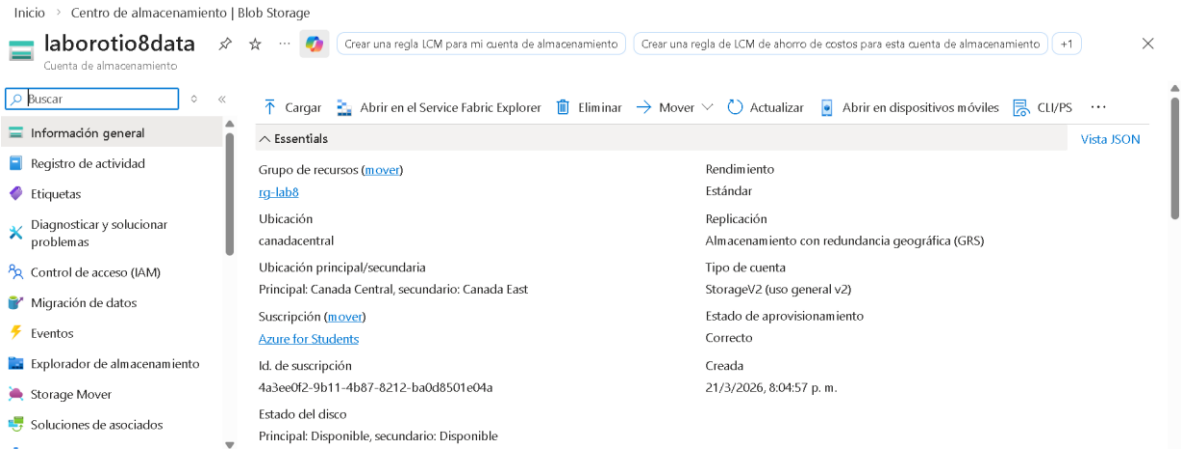
```
C:\Users\pacoa\Desktop\Laboratorio-8-ARSW\infra>az account show
{
  "environmentName": "AzureCloud",
  "homeTenantId": "50640584-2a40-4216-a84b-9b3ee0f3f6cf",
  "id": "4a3ee0f2-9b11-4b87-8212-ba0d8501e04a",
  "isDefault": true,
  "managedByTenants": [],
  "name": "Azure for Students",
  "state": "Enabled",
  "tenantId": "50640584-2a40-4216-a84b-9b3ee0f3f6cf",
  "user": {
    "name": "oscar.sporras@mail.escuelaing.edu.co",
    "type": "user"
  }
}
```

```
C:\Users\pacoa\Desktop\Laboratorio-8-ARSW\infra>terraform init -reconfigure -backend-config=backend.hcl
Initializing the backend...

Successfully configured the backend "azurerm"! Terraform will automatically
use this backend unless the backend configuration changes.
Initializing modules...
Initializing provider plugins...
- Finding hashicorp/azurerm versions matching "~> 4.0"...
- Installing hashicorp/azurerm v4.65.0...
- Installed hashicorp/azurerm v4.65.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```





terraform init -reconfigure -backend-config=backend.hcl

```
C:\Users\pacoa\Desktop\Laboratorio-8-ARSW\infra>terraform init -
reconfigure -backend-config=backend.hcl
Initializing the backend...

Successfully configured the backend "azurerm"! Terraform will au
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Initializing modules...
Initializing provider plugins...
- Finding hashicorp/azurerm versions matching "~> 4.0"...
- Installing hashicorp/azurerm v4.65.0...
- Installed hashicorp/azurerm v4.65.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record
the provider
selections it made above. Include this file in your version cont
rol repository
so that Terraform can guarantee to make the same selections by d
efault when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform
 plan" to see
any changes that are required for your infrastructure. All Terra
form commands
should now work.
```

terraform fmt -recursive

```
C:\Users\pacoa\Desktop\Laboratorio-8-ARSW\infra>terraform fmt -r
ecursive
env\dev.tfvars
```

terraform validate

```
C:\Users\pacoa\Desktop\Laboratorio-8-ARSW\infra>terraform valida
te
Success! The configuration is valid.
```

terraform plan -var-file=env/dev.tfvars -out plan.tfplan

```
# module.vnet.azurerm_virtual_network.vnet will be created
+ resource "azurerm_virtual_network" "vnet" {
    + address_space                  = [
        + "10.10.0.0/16",
      ]
    + dns_servers                    = (known after apply)
    + guid                           = (known after apply)
    + id                             = (known after apply)
    + location                       = "eastus2"
    + name                           = "lab8-vnet"
    + private_endpoint_vnet_policies = "Disabled"
    + resource_group_name            = "lab8-rg"
    + subnet                         = (known after apply)
    + tags                           = {
        + "course"  = "ARSW"
        + "env"     = "dev"
        + "expires" = "2025-12-31"
        + "owner"   = "alias"
      }
  }

lan: 18 to add, 0 to change, 1 to destroy.

hanges to Outputs:
 + lb_public_ip        = (known after apply)


aved the plan to: plan.tfplan
```

terraform apply "plan.tfplan"

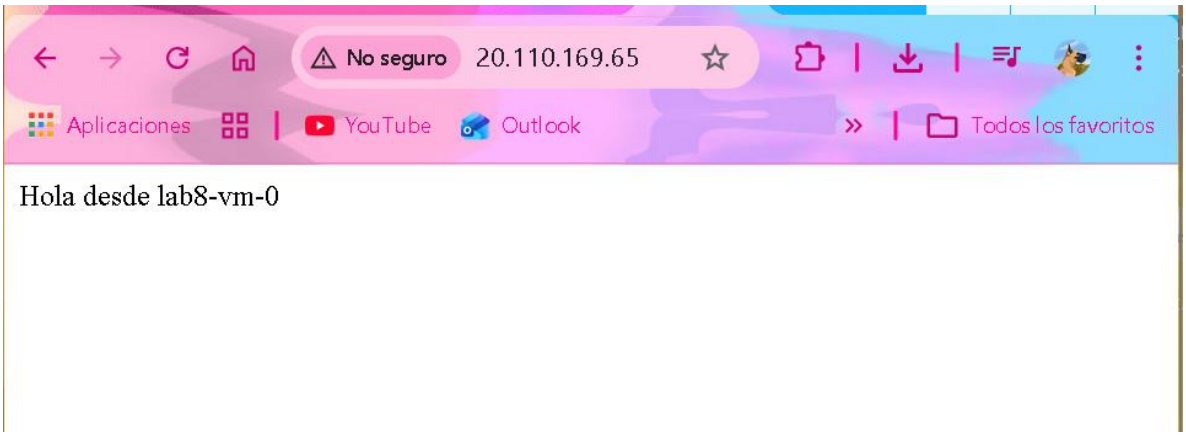
```
module.compute.azurerm_linux_virtual_machine.vm[1]: Still creati
ng... [00m50s elapsed]
module.compute.azurerm_linux_virtual_machine.vm[0]: Still creati
ng... [00m50s elapsed]
module.compute.azurerm_linux_virtual_machine.vm[0]: Still creati
ng... [01m00s elapsed]
module.compute.azurerm_linux_virtual_machine.vm[1]: Still creati
ng... [01m00s elapsed]
module.compute.azurerm_linux_virtual_machine.vm[0]: Creation com
plete after 1m4s [id=/subscriptions/4a3ee0f2-9b11-4b87-8212-ba0d
8501e04a/resourceGroups/lab8-rg/providers/Microsoft.Compute/virt
ualMachines/lab8-vm-0]
module.compute.azurerm_linux_virtual_machine.vm[1]: Creation com
plete after 1m8s [id=/subscriptions/4a3ee0f2-9b11-4b87-8212-ba0d
8501e04a/resourceGroups/lab8-rg/providers/Microsoft.Compute/virt
ualMachines/lab8-vm-1]
Releasing state lock. This may take a few moments...

Apply complete! Resources: 18 added, 0 changed, 1 destroyed.

Outputs:

lb_public_ip = "20.110.169.65"
resource_group_name = "lab8-rg"
vm_names = [
  "lab8-vm-0",
  "lab8-vm-1",
]
```

La infraestructura fue desplegada exitosamente, generando:

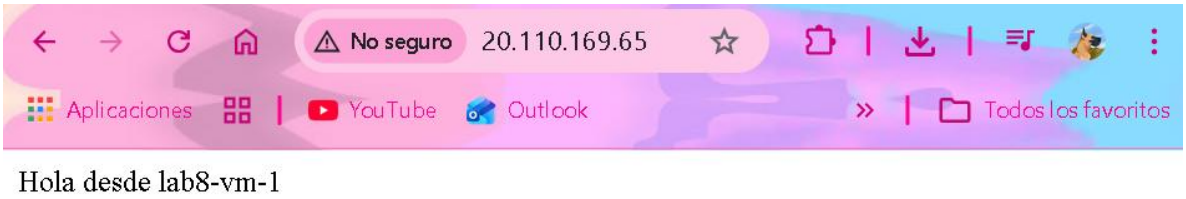
- Dos instancias virtuales activas
- Balanceo de carga funcional
- Acceso público mediante la IP:

20.110.169.65

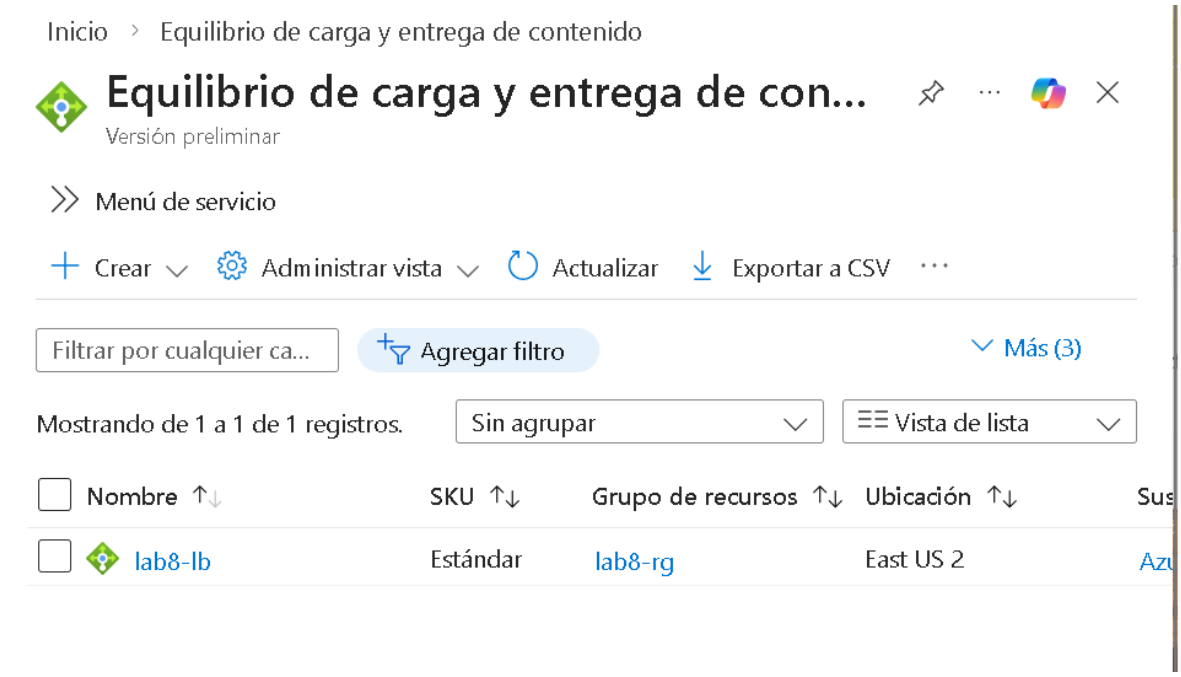
Se verificó el funcionamiento accediendo vía navegador, confirmando la correcta distribución del tráfico.

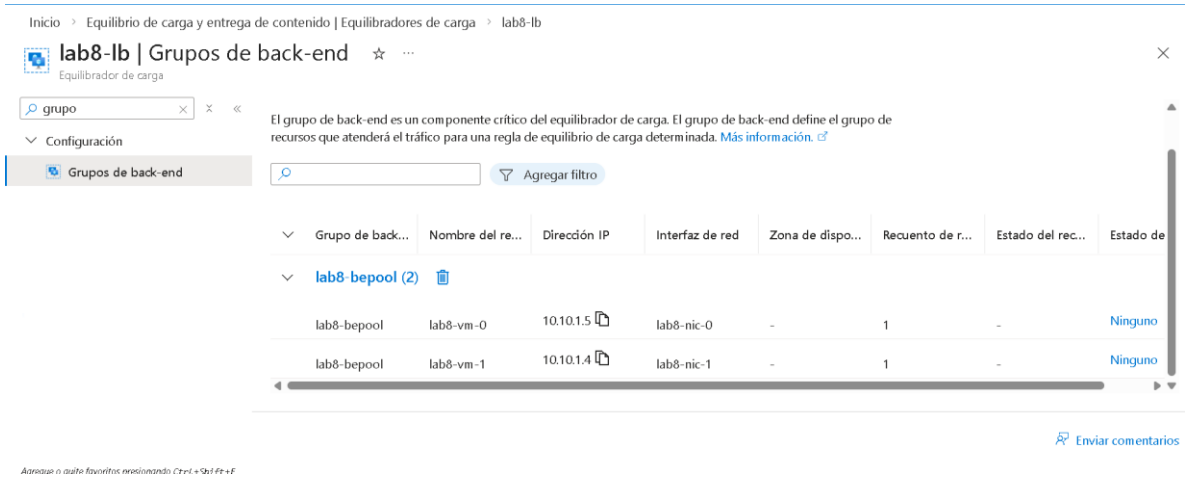


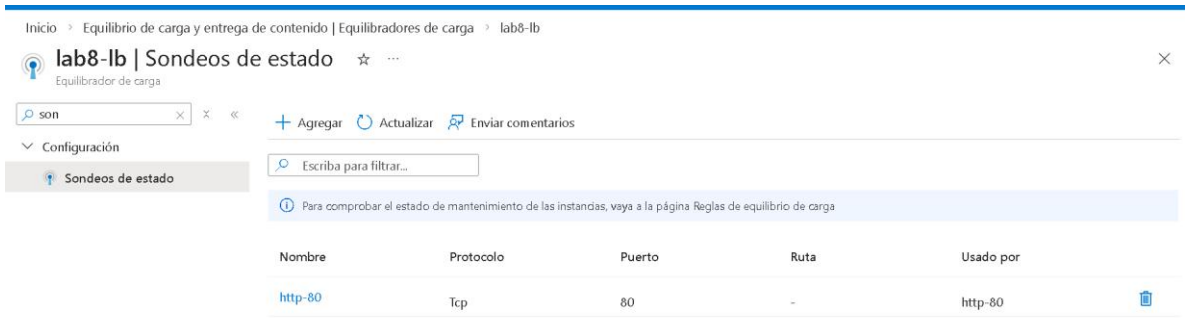
Luego recargar la pagina y se muestra un cambio lo que significa que esta funcionando el Load Balancer.




```
C:\Users\pacoa\Desktop\Laboratorio-8-ARSW\infra>terraform output

lb_public_ip = "20.110.169.65"
resource_group_name = "lab8-rg"
vm_names = [
  "lab8-vm-0",
  "lab8-vm-1",
]
```







Comprobante de bakcend remoto

```
C:\Users\pacoa\Desktop\Laboratorio-8-ARSW\infra>terraform state
list
azurerm_resource_group.rg
module.compute.azurerm_linux_virtual_machine.vm[0]
module.compute.azurerm_linux_virtual_machine.vm[1]
module.compute.azurerm_network_interface.nic[0]
module.compute.azurerm_network_interface.nic[1]
module.lb.azurerm_lb.lb
module.lb.azurerm_lb_backend_address_pool.bepool
module.lb.azurerm_lb_probe.probe
module.lb.azurerm_lb_rule.rule
module.lb.azurerm_network_interface_backend_address_pool_associa
tion.assoc[0]
module.lb.azurerm_network_interface_backend_address_pool_associa
tion.assoc[1]
module.lb.azurerm_network_interface_security_group_association.n
sg_assoc[0]
module.lb.azurerm_network_interface_security_group_association.n
sg_assoc[1]
module.lb.azurerm_network_security_group.nsg
module.lb.azurerm_public_ip.pip
module.vnet.azurerm_subnet.mgmt
module.vnet.azurerm_subnet.web
module.vnet.azurerm_virtual_network.vnet
```

```
C:\Users\pacoa\Desktop\Laboratorio-8-ARSW\infra>terraform output

lb_public_ip = "20.110.169.65"
resource_group_name = "lab8-rg"
vm_names = [
  "lab8-vm-0",
  "lab8-vm-1",
]
```

```
C:\Users\pacoa\Desktop\Laboratorio-8-ARSW\infra>terraform output
 -raw lb_public_ip
20.110.169.65
```

```
pacoa@LAPTOP-DO3ENF55 MINGW64 ~/Desktop/Laboratorio-8-ARSW/infra (develop)
$ curl http://$(terraform output -raw lb_public_ip)
Hola desde lab8-vm-0
```

```
pacoa@LAPTOP-D03ENF55 MINGW64 ~/Desktop/Laboratorio-8-ARSW/infra (develop)
$ curl http://20.110.169.65
Hola desde lab8-vm-0

pacoa@LAPTOP-D03ENF55 MINGW64 ~/Desktop/Laboratorio-8-ARSW/infra (develop)
$ curl http://20.110.169.65
Hola desde lab8-vm-1

pacoa@LAPTOP-D03ENF55 MINGW64 ~/Desktop/Laboratorio-8-ARSW/infra (develop)
$
```

```
C:\Users\pacoa\Desktop\Laboratorio-8-ARSW\infra>az ad app creat
 --display-name "lab8-terraform"
{
  "@odata.context": "https://graph.microsoft.com/v1.0/$metadata
applications/$entity",
  "addIns": [],
  "api": {
    "acceptMappedClaims": null,
    "knownClientApplications": [],
    "oauth2PermissionScopes": [],
    "preAuthorizedApplications": [],
    "requestedAccessTokenVersion": null
  },
  "appId": "08a56082-3cd3-468b-b472-24f38abcaa73",
```

```
C:\Users\pacoa\Desktop\Laboratorio-8-ARSW\infra>az ad sp create
--id 08a56082-3cd3-468b-b472-24f38abcaa73
{
  "@odata.context": "https://graph.microsoft.com/v1.0/$metadata#
servicePrincipals/$entity",
  "accountEnabled": true,
  "addIns": [],
  "alternativeNames": [],
  "appDescription": null,
  "appDisplayName": "lab8-terraform",
  "appId": "08a56082-3cd3-468b-b472-24f38abcaa73",
  "appOwnerOrganizationId": "50640584-2a40-4216-a84b-9b3ee0f3f6c
f"
```

```
PS C:\Users\pacoa> az role assignment create --assignee 08a56082
-3cd3-468b-b472-24f38abcaa73 --role Contributor --scope /subscri
ptions/4a3ee0f2-9b11-4b87-8212-ba0d8501e04a/resourceGroups/lab8-
rg
{
  "condition": null,
  "conditionVersion": null,
  "createdBy": null,
  "createdOn": "2026-03-22T05:13:36.671816+00:00",
  "delegatedManagedIdentityResourceId": null,
  "description": null,
  "id": "/subscriptions/4a3ee0f2-9b11-4b87-8212-ba0d8501e04a/res
ourceGroups/lab8-rg/providers/Microsoft.Authorization/roleAssign
ments/54b0c376-8cd6-4975-9390-fa8a39d15740",
  "name": "54b0c376-8cd6-4975-9390-fa8a39d15740",
  "principalId": "99394f7c-b8c4-49bc-8aa5-08906dc61df3",
  "principalType": "ServicePrincipal",
  "resourceGroup": "lab8-rg",
  "roleDefinitionId": "/subscriptions/4a3ee0f2-9b11-4b87-8212-ba
0d8501e04a/providers/Microsoft.Authorization/roleDefinitions/b24
988ac-6180-42a0-ab88-20f7382dd24c",
  "scope": "/subscriptions/4a3ee0f2-9b11-4b87-8212-ba0d8501e04a/
resourceGroups/lab8-rg",
  "type": "Microsoft.Authorization/roleAssignments",
  "updatedBy": "56ebbdd3-b301-426a-b735-a6737f28cfaf",
  "updatedOn": "2026-03-22T05:13:36.974825+00:00"
}
```

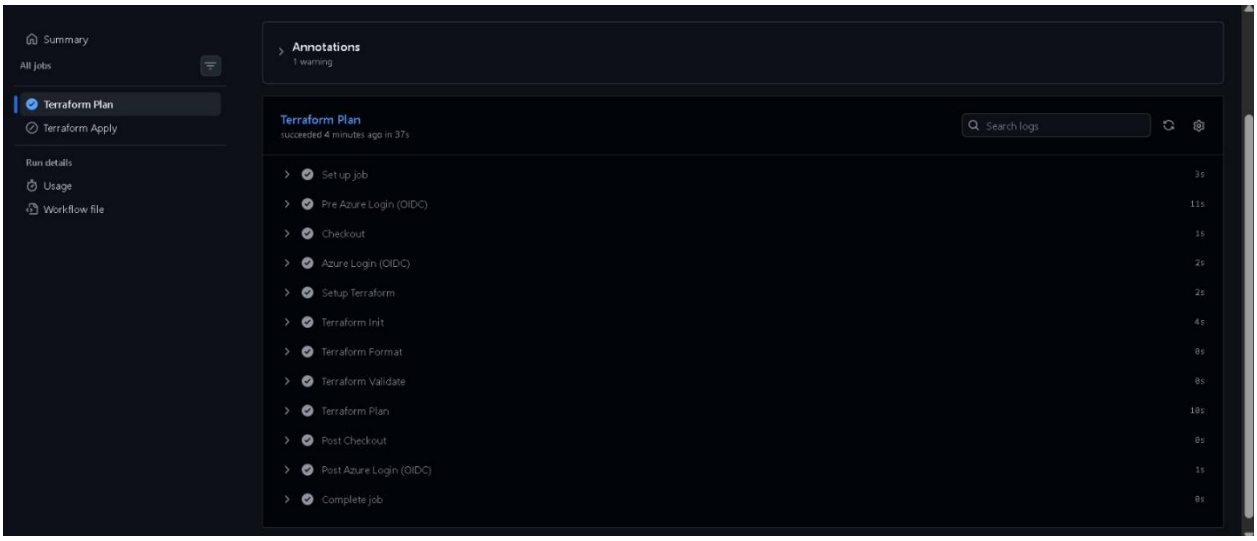
```
pacoa@LAPTOP-DO3ENF55 MINGW64 ~/Desktop/Laboratorio-8-ARSW (develop)
$ az ad app federated-credential create --id 08a56082-3cd3-468b-b472-24f38abcaa7
3 --parameters '{"name":"github-pr","issuer":"https://token.actions.githubuserco
ntent.com","subject":"repo:D1052023/Laboratorio-8-ARSW:pull_request","audiences"
:["api://AzureADTokenExchange"]}'
{
  "@odata.context": "https://graph.microsoft.com/v1.0/$metadata#applications('a1
c35ede-3eba-4b81-8907-8118f0de0f1b')/federatedIdentityCredentials/$entity",
  "audiences": [
    "api://AzureADTokenExchange"
  ],
  "description": null,
  "id": "be2493cb-904d-4dec-9cde-b3f7aa628a26",
  "issuer": "https://token.actions.githubusercontent.com",
  "name": "github-pr",
  "subject": "repo:D1052023/Laboratorio-8-ARSW:pull_request"
}
```

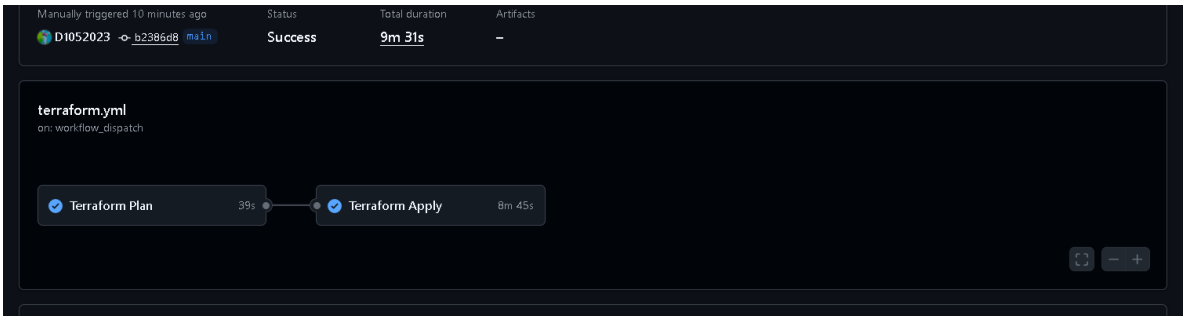
```
pacoa@LAPTOP-D03ENF55 MINGW64 ~/Desktop/Laboratorio-8-ARSW (develop)
$ az ad app federated-credential create --id 08a56082-3cd3-468b-b472-24f38abcaa
3 --parameters '{"name":"github-main","issuer":"https://token.actions.githubuse
content.com","subject":"repo:D1052023/Laboratorio-8-ARSW:ref:refs/heads/main","
udiences":["api://AzureADTokenExchange"]}'
{
  "@odata.context": "https://graph.microsoft.com/v1.0/$metadata#applications('a
c35ede-3eba-4b81-8907-8118f0de0f1b')/federatedIdentityCredentials/$entity",
  "audiences": [
    "api://AzureADTokenExchange"
  ],
  "description": null,
  "id": "fd412211-65ab-42a4-8323-38116cf046c2",
  "issuer": "https://token.actions.githubusercontent.com",
  "name": "github-main",
  "subject": "repo:D1052023/Laboratorio-8-ARSW:ref:refs/heads/main"
}
```

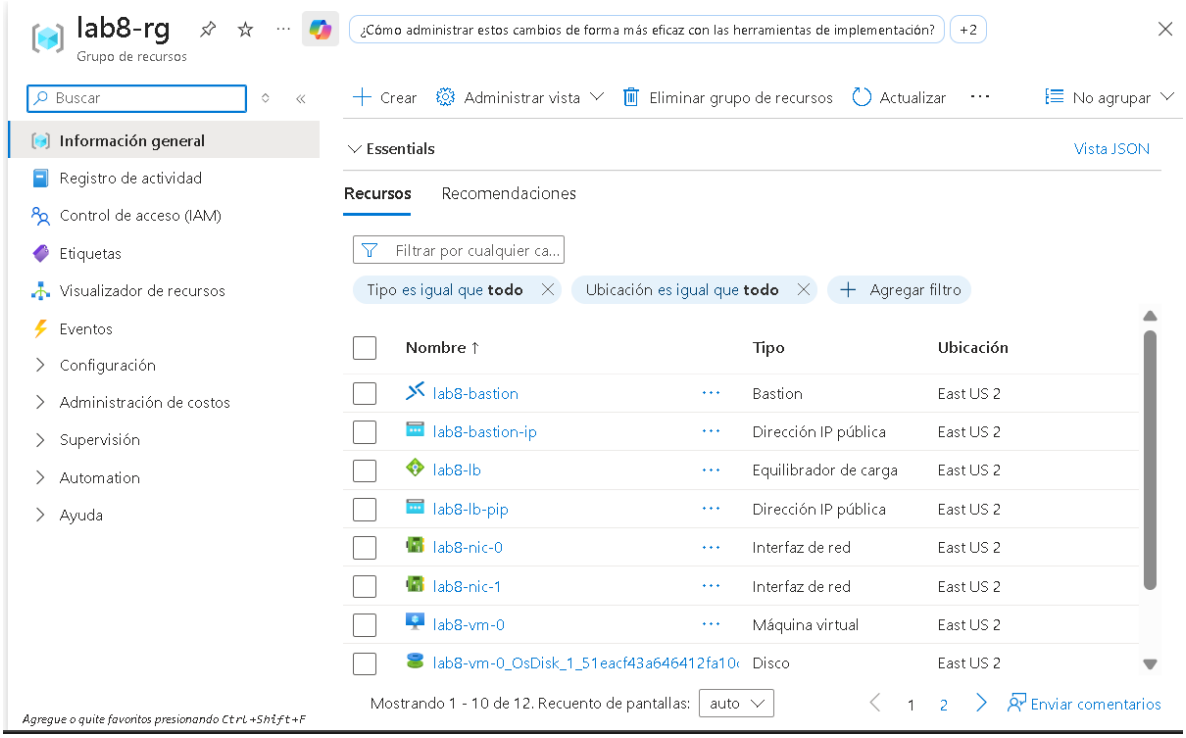
C:\Windows\System32\cmd.e

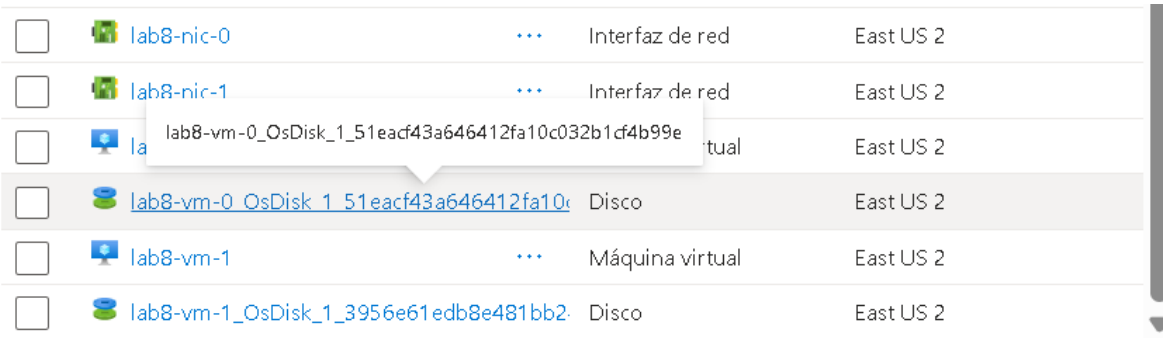
```
:\Users\pacoa\Desktop\Laboratorio-8-ARSW\infra>az role assignme
t create --assignee 08a56082-3cd3-468b-b472-24f38abcaa73 --role
"Storage Blob Data Contributor" --scope /subscriptions/4a3ee0f2
9b11-4b87-8212-ba0d8501e04a/resourceGroups/rg-lab8/providers/Mi
rosoft.Storage/storageAccounts/laborotio8data

 "condition": null,
 "conditionVersion": null,
 "createdBy": null,
 "createdOn": "2026-03-22T17:12:34.949107+00:00",
 "delegatedManagedIdentityResourceId": null,
 "description": null,
 "id": "/subscriptions/4a3ee0f2-9b11-4b87-8212-ba0d8501e04a/res
urceGroups/rg-lab8/providers/Microsoft.Storage/storageAccounts/
aborotio8data/providers/Microsoft.Authorization/roleAssignments
09e4a082-5851-4f94-870a-65f5df3d442e",
 "name": "09e4a082-5851-4f94-870a-65f5df3d442e",
 "principalId": "99394f7c-b8c4-49bc-8aa5-08906dc61df3",
 "principalType": "ServicePrincipal",
 "resourceGroup": "rg-lab8",
 "roleDefinitionId": "/subscriptions/4a3ee0f2-9b11-4b87-8212-ba
d8501e04a/providers/Microsoft.Authorization/roleDefinitions/ba9
f5b4-2d11-453d-a403-e96b0029c9fe",
 "scope": "/subscriptions/4a3ee0f2-9b11-4b87-8212-ba0d8501e04a/
esourceGroups/rg-lab8/providers/Microsoft.Storage/storageAccoun
s/laborotio8data",
 "type": "Microsoft.Authorization/roleAssignments",
 "updatedBy": "56ebbdd3-b301-426a-b735-a6737f28cfaf",
 "updatedOn": "2026-03-22T17:12:35.248107+00:00"


:\Users\pacoa\Desktop\Laboratorio-8-ARSW\infra>
```

```
C:\Users\pacoa\Desktop\Laboratorio-8-ARSW\infra>az storage blob
list --account-name laborotio8data --container-name lab8arsw --a
uth-mode login
[
  {
    "container": "lab8arsw",
    "content": "",
    "deleted": null,
    "encryptedMetadata": null,
    "encryptionKeySha256": null,
    "encryptionScope": null,
    "hasLegalHold": null,
    "hasVersionsOnly": null,
    "immutabilityPolicy": {
      "expiryTime": null,
      "policyMode": null
    },
    "isAppendBlobSealed": null,
    "isCurrentVersion": null,
    "lastAccessedOn": null,
    "metadata": {},
    "name": "lab8arsw.tfstate",
    "objectReplicationDestinationPolicy": null,
    "objectReplicationSourceProperties": [],
    "properties": {
      "appendBlobCommittedBlockCount": null,
      "blobTier": "Hot",
      "blobTierChangeTime": null,
      "blobTierInferred": true,
      "blobType": "BlockBlob",
```




Manually triggered 10 minutes ago
D1052023 b2386d8 main
Status
Success
Total duration
9m 31s
Artifacts
–
terraform.yml
on: workflow_dispatch
Terraform Plan 39s
Terraform Apply 8m 45s


lab8-rg
Grupo de recursos
¿Cómo administrar estos cambios de forma más eficaz con las herramientas de implementación?
+2
Buscar
Información general
Registro de actividad
Control de acceso (IAM)
Etiquetas
Visualizador de recursos
Eventos
Configuración
Administración de costos
Supervisión
Automation
Ayuda
Crear
Administrar vista
Eliminar grupo de recursos
Actualizar
No agrupar
Essentials
Vista JSON
Recursos
Recomendaciones
Filtrar por cualquier ca...
Tipo es igual que todo
Ubicación es igual que todo
Agregar filtro
Nombre
Tipo
Ubicación
lab8-bastion Bastion East US 2
lab8-bastion-ip Dirección IP pública East US 2
lab8-lb Equilibrador de carga East US 2
lab8-lb-pip Dirección IP pública East US 2
lab8-nic-0 Interfaz de red East US 2
lab8-nic-1 Interfaz de red East US 2
lab8-vm-0 Máquina virtual East US 2
lab8-vm-0_OsDisk_1_51eacf43a646412fa10 Disco East US 2
Mostrando 1 - 10 de 12. Recuento de pantallas: auto
1 2
Enviar comentarios
Agregue o quite favoritos presionando Ctrl+Shift+F


lab8-nic-0 Interfaz de red East US 2
lab8-nic-1 Interfaz de red East US 2
lab8-vm-0_OsDisk_1_51eacf43a646412fa10c032b1cf4b99e
lab8-vm-0_OsDisk_1_51eacf43a646412fa10 Disco East US 2
lab8-vm-1 Máquina virtual East US 2
lab8-vm-1_OsDisk_1_3956e61edb8e481bb2 Disco East US 2

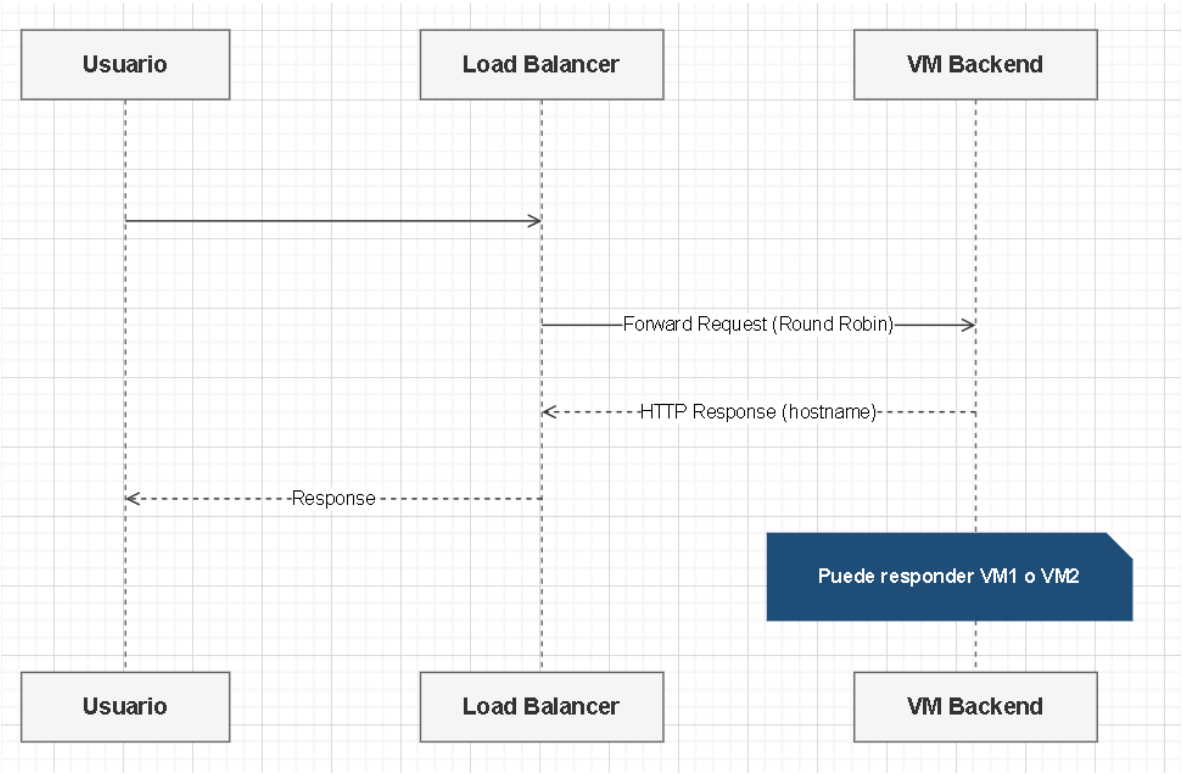
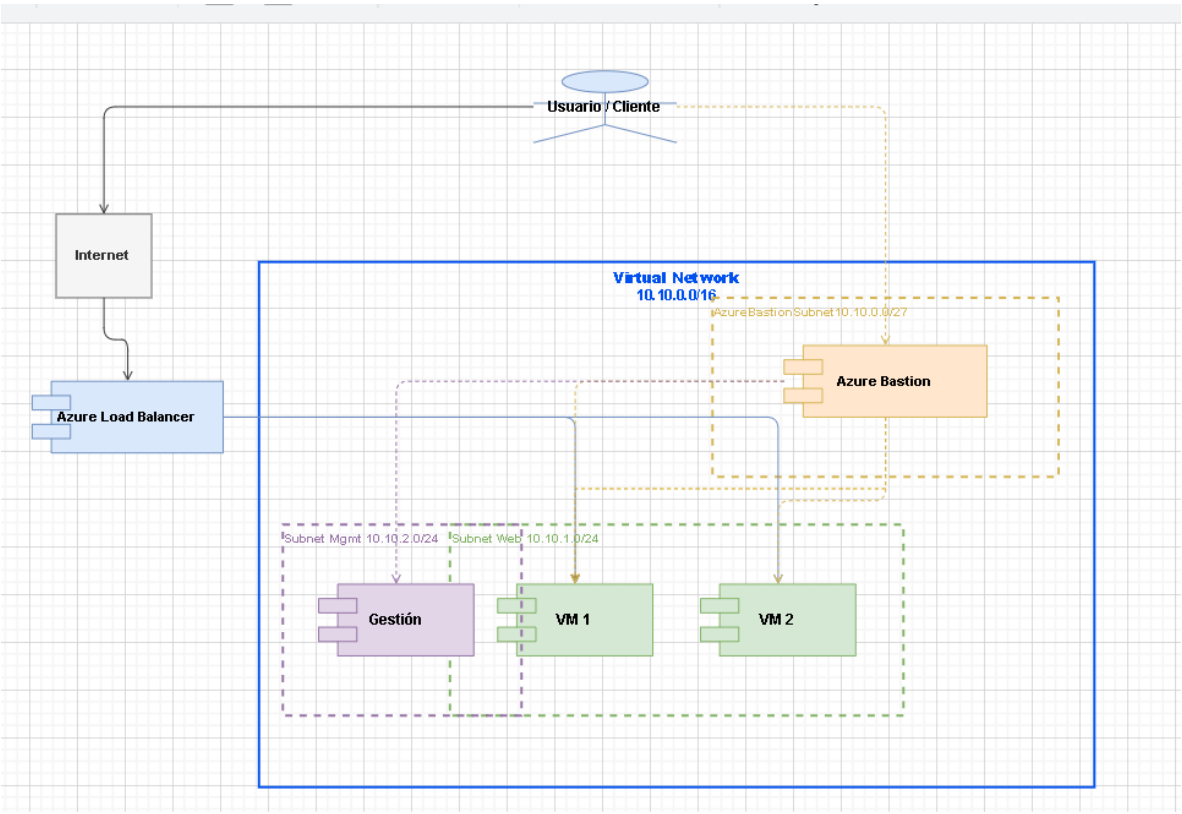
C:\Windows\System32\cmd.e

```
/subscriptions/4a3ee0f2-9b11-4b87-8212-...work/publicIPAddress
s/lab8-bastion-ip, 00m10s elapsed]
odule.vnet.azurerm_subnet.bastion: Still destroying... [id=/sub
criptions/4a3ee0f2-9b11-4b87-8212-...s/lab8-vnet/subnets/AzureB
stionSubnet, 00m10s elapsed]
odule.vnet.azurerm_public_ip.bastion_ip: Destruction complete a
ter 11s
odule.vnet.azurerm_subnet.bastion: Destruction complete after 1
s
odule.vnet.azurerm_virtual_network.vnet: Destroying... [id=/sub
criptions/4a3ee0f2-9b11-4b87-8212-ba0d8501e04a/resourceGroups/l
b8-rg/providers/Microsoft.Network/virtualNetworks/lab8-vnet]
odule.vnet.azurerm_virtual_network.vnet: Still destroying... [i
/subscriptions/4a3ee0f2-9b11-4b87-8212-...soft.Network/virtual
etworks/lab8-vnet, 00m10s elapsed]
odule.vnet.azurerm_virtual_network.vnet: Destruction complete a
ter 11s
zurerm_resource_group.rg: Destroying... [id=/subscriptions/4a3e
0f2-9b11-4b87-8212-ba0d8501e04a/resourceGroups/lab8-rg]
zurerm_resource_group.rg: Still destroying... [id=/subscription
/4a3ee0f2-9b11-4b87-8212-ba0d8501e04a/resourceGroups/lab8-rg, 0
m10s elapsed]
zurerm_resource_group.rg: Still destroying... [id=/subscription
/4a3ee0f2-9b11-4b87-8212-ba0d8501e04a/resourceGroups/lab8-rg, 0
m20s elapsed]
zurerm_resource_group.rg: Destruction complete after 22s
eleasing state lock. This may take a few moments...

estroy complete! Resources: 22 destroyed.

\Users\pacoa\Desktop\Laboratorio-8-ARSW\infra>
\Users\pacoa\Desktop\Laboratorio-8-ARSW\infra>
```

```
# azurerm_consumption_budget_resource_group.budget will be created
+ resource "azurerm_consumption_budget_resource_group" "budget" {
    + amount            = 10
    + etag              = (known after apply)
    + id                = (known after apply)
    + name              = "lab8-budget"
    + resource_group_id = "/subscriptions/***/resourceGroups/lab8-rg"
    + time_grain        = "Monthly"

    + notification {
        + contact_emails = [
            + "oscar.sporras@mail.escuelaing.edu.co",
          ]
        + contact_groups = []
        + contact_roles  = []
        + enabled        = true
        + operator       = "GreaterThan"
        + threshold      = 80
        + threshold_type = "Actual"
      }

    + time_period {
        + end_date   = (known after apply)
        + start_date = "2026-03-01T00:00:00Z"
      }
  }

Plan: 1 to add, 0 to change, 0 to destroy.
```

Diagramas

Secuencia



Componentes